## What Is Ensemble Learning?

Ensemble learning refers to a machine learning approach where several models are trained to address a common problem, and their predictions are combined to enhance the overall performance. The idea behind ensemble learning is that by combining multiple models, each with its strengths and weaknesses, the ensemble can achieve better results than any single model alone. Ensemble learning can be applied to various machine learning tasks, including classification, regression, and clustering.

## Ensemble Techniques

Ensemble techniques in machine learning involve combining multiple models to improve performance. One common ensemble technique is bagging, which uses bootstrap sampling to create multiple datasets from the original data and trains a model on each dataset. Another technique is boosting, which trains models sequentially, each focusing on the previous models' mistakes. Random forests are a popular ensemble method that uses decision trees as base learners and combines their predictions to make a final prediction. Ensemble techniques are effective because they reduce overfitting and improve generalization, leading to more robust models.

## Simple Ensemble Techniques

Simple ensemble techniques combine predictions from multiple models to produce a final prediction. These techniques are straightforward to implement and can often improve performance compared to individual models.

In **voting**, multiple models are trained independently on the same dataset, and their predictions are combined by **voting** in the case of classification tasks, or by **averaging** in the case of regression tasks.

### *Max Voting*

In this technique, the final prediction is the most frequent prediction among the base models. For example, if three base models predict the classes A, B, and A for a given sample, the final prediction using max voting would be class A, as it appears more frequently.

### *Averaging*

Averaging involves taking the average of predictions from multiple models. This can be particularly useful for regression problems, where the final prediction is the mean of predictions from all models. For classification, averaging can be applied to the predicted probabilities for a more confident prediction.

### *Example:*

You can train three models (e.g., logistic regression, decision tree, and random forest) on a dataset and combine their predictions by hard voting. The final prediction is based on the majority vote.

### *Advantages of Voting:*

- Simple to implement and interpret.
- Can improve accuracy by combining diverse models.
- Works well when the base models are fairly strong and complementary. It's better to use those model whose accuracy more than 50%.

### *Stacking*

Example: Let's say we have 4 input features (x1,x2,x3,x4) and 1 output feature (y), And we have 8 records in the dataset

For stacking, let's say the model m1 is trained. Let the k fold size be 2. Which means that for first iteration it will take first 6 records for training and then give the output for the last 2 records. In the second iteration it will take first 4 records, and last 2 records for training and give output for 5th and 6th record. Similarly 2 more iterations will happen. And the prediction by the model on each of the k fold dataset be p1, p2, p3, p4, p5, p6, p7 and p8.

Now similarly other lets say 2 models will be trained in a similar way and let its prediction be p9,p10,p11,p12,p13,p14,p15,p16 and p17,p18,p19,p20,p21,p22,p23,p24.

Here note that p1, p9 and p17 are predictions made by different model on the same record and similarly for p2, p10, p18 and so on and so forth

Now this data will be sent to the meta model to learn the relationship between the models prediction and the actual y value.

One such row for the dataset which will be sent to the meta model will look like (p1, p9, p17 as input feature and the output as the actual output value of the record on which the each of the 3 model has given its prediction)

### *Key Characteristics of Stacking:*

- **Training Data**: The original dataset is divided into **k-folds** for cross-validation:
- Each base model is trained on k−1 folds and makes predictions on the remaining fold. This process is repeated k times, with each fold being used as a validation set once.
- Predictions from each base model across all folds are collected and used to train the meta-model.
- **Meta-Model**: Trained on predictions from the base models across all folds. This meta-model learns to combine the base models' predictions to make the final prediction.

## *Bagging*

Bagging (Bootstrap Aggregating) is an ensemble learning technique designed to improve the accuracy and stability of machine learning algorithms. It involves the following steps:

Data Sampling: Creating multiple subsets of the training dataset using bootstrap sampling (random sampling with replacement).

Model Training: raining a separate model on each subset of the data.

Aggregation: Combining the predictions from all individual models (averaged for regression or majority voting for classification) to produce the final output.

Example:

Suppose you have a dataset with 1,000 emails to classify as spam or not spam. You create five subsets of 200 emails each (with replacement) and train five decision tree classifiers. To make a prediction for a new email, you let these five decision trees vote, and the majority decision is the final prediction

### *Key Benefits:*

Reduces Variance: By averaging multiple predictions, bagging reduces the variance of the model and helps prevent overfitting.

Improves Accuracy: Combining multiple models usually leads to better performance than individual models.

Example of Bagging Algorithms:

Random Forests (an extension of bagging applied to decision trees)

### Random Forests

A random forest is a machine learning technique that's used to solve regression and classification problems. It utilizes ensemble learning (bagging), which is a technique that combines many classifiers to provide solutions to complex problems.

A random forest algorithm consists of many decision trees. The 'forest' generated by the random forest algorithm is trained through bagging or bootstrap aggregating. Bagging is an ensemble meta-algorithm that improves the accuracy of machine learning algorithms.

The (random forest) algorithm establishes the outcome based on the predictions of the decision trees. It predicts by taking the average or mean of the output from various trees. Increasing the number of trees increases the precision of the outcome.

Features of a Random Forest Algorithm

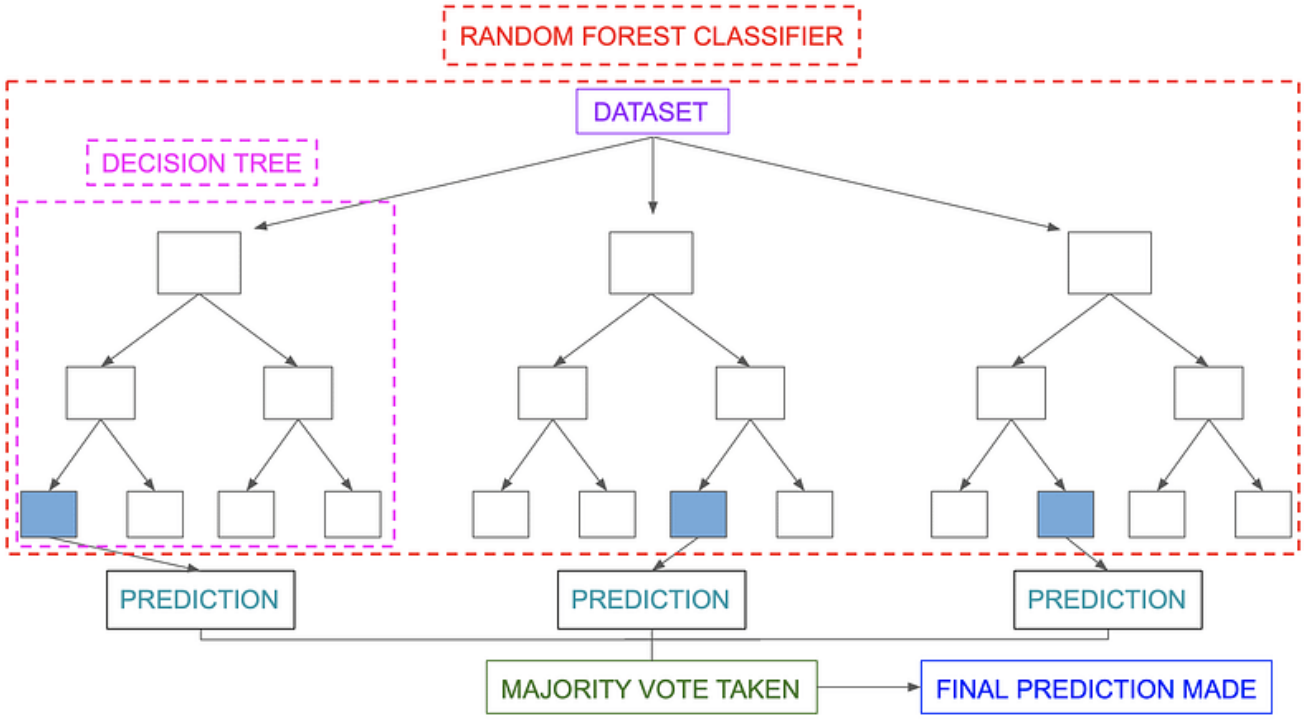
- It's more accurate than the decision tree algorithm.
- It provides an effective way of handling missing data.
- It can produce a reasonable prediction without hyper-parameter tuning.
- It solves the issue of overfitting in decision trees.

- In every random forest tree, a subset of features is selected randomly at the node's splitting point.

Classification in random forests

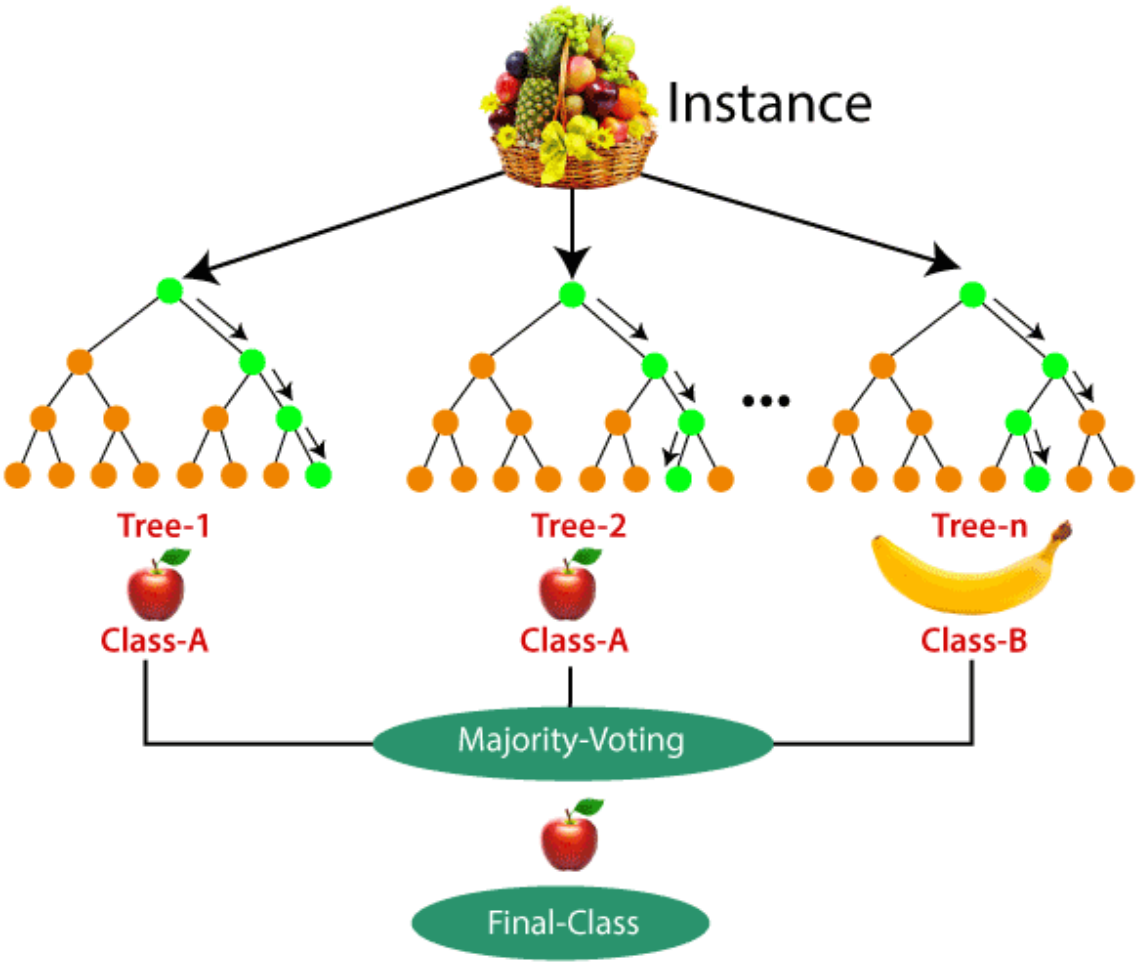
Classification in random forests employs an ensemble methodology to attain the outcome. The training data is fed to train various decision trees. This dataset consists of observations and features that will be selected randomly during the splitting of nodes.

A rain forest system relies on various decision trees. Every decision tree consists of decision nodes, leaf nodes, and a root node. The leaf node of each tree is the final output produced by that specific decision tree. The selection of the final output follows the majority-voting system. In this case, the output chosen by the majority of the decision trees becomes the final output of the rain forest system. The diagram below shows a simple random forest classifier.




Let's take an example of a training dataset consisting of various fruits such as bananas, apples, pineapples, and mangoes. The random forest classifier divides this dataset into subsets. These subsets are given to every decision tree in the random forest system. Each decision tree produces its specific output. For example, the prediction for trees 1 and 2 is apple.

Another decision tree (n) has predicted banana as the outcome. The random forest classifier collects the majority voting to provide the final prediction. The majority of the decision trees have chosen apple as their prediction. This makes the classifier choose apple as the final prediction.




**Regression in random forests**

Regression is the other task performed by a random forest algorithm. A random forest regression follows the concept of simple regression. Values of dependent (features) and independent variables are passed in the random forest model.

We can run random forest regressions in various programs such as JAVA, R, and python. In a random forest regression, each tree produces a specific prediction. The mean prediction of the individual trees is the output of the regression. This is contrary to random forest classification, whose output is determined by the mode of the decision trees' class.

## *Boosting*

Boosting is another ensemble learning technique that focuses on creating a strong model by combining several weak models. It involves the following steps:

Sequential Training: Training models sequentially, each one trying to correct the errors made by the previous models.

Weight Adjustment: Each instance in the training set is weighted. Initially, all instances have equal weights. After each model is trained, the weights of misclassified instances are increased so that the next model focuses more on difficult cases.

Model Combination: Combining the predictions from all models to produce the final output, typically by weighted voting or weighted averaging.

Key Benefits:

Reduces Bias: By focusing on hard-to-classify instances, boosting reduces bias and improves the overall model accuracy.

Produces Strong Predictors: Combining weak learners leads to a strong predictive model.

Example of Boosting Algorithms:

AdaBoost, Gradient Boosting Machines (GBM), XGBoost, LightGBM